

Internship project

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Charge Pump PLL Design using Cadence Virtuoso

Abstract

This report documents the complete transistor-level design and verification of a Charge Pump PLL.

Project Specifications

- Technology: GPDK090
- VDD: 1.2 V
- PMOS and NMOS Width - 120nm
- PMOS and NMOS length – 100nm
- Current starving NMOS and PMOS width – 240nm
- Architecture: Charge Pump PLL
- Icp: 30 uA
- Frequency Divider: /8

What is a PLL?

A **Phase Locked Loop (PLL)** is a closed-loop feedback system that synchronizes the phase and frequency of an output signal with a reference signal. It continuously compares the reference clock with the feedback clock and adjusts the Voltage Controlled Oscillator (VCO) until both signals operate at the same frequency and phase.

Why do we need PLL?

PLLs are widely used whenever a stable and accurate clock signal is required. They generate high-frequency clocks from a low-frequency reference, reduce clock skew, recover clock signals from transmitted

data, and maintain synchronization in communication and digital systems.

Applications

- Frequency Synthesizers
- Clock Generation
- Clock Recovery (CDR)
- Microprocessors and SoCs
- Wireless Communication Systems
- High-Speed Serial Interfaces
- Phase Modulation/Demodulation
- Radar Systems
- Satellite Communication
- Digital Communication Systems

Mathematics

Phase Error: $\varphi_e = \varphi_{ref} - \varphi_{fb}$

Lock Condition:

$F_{ref} = F_{fb} = 301.25 \text{ MHz}$

Divider:

$F_{fb} = F_{vco} / N$

$F_{vco} = N * f_{ref} \quad \{ N = \text{frequency divide by divider} \}$

$2.41 \text{ GHz} = 301.25 * 8$

Charge Pump:

Average Charge Pump Current at Lock:

$$I_{avg} \approx 0 \mu A$$

because $t_{UP} \approx t_{DOWN}$.

Loop Filter:

$$Z(s) = R + 1/(sC1)$$

$$\omega_p = 1/(RC2)$$

$$\omega_p = 5.68 * 10^7 \text{ rad/s}$$

$$K_{vco} = \Delta f / \Delta V_c = (2.504 - 2.332) / (1.1 - 0.1) = 0.172 \text{ GHz/V} = 172 \text{ MHz/V}$$

VCO:

$$f_{vco} = f_{free} + K_{vco} \times V_c$$

The free-running frequency of the VCO is approximately **2.33 GHz** (measured near the minimum control voltage of 0.1 V).

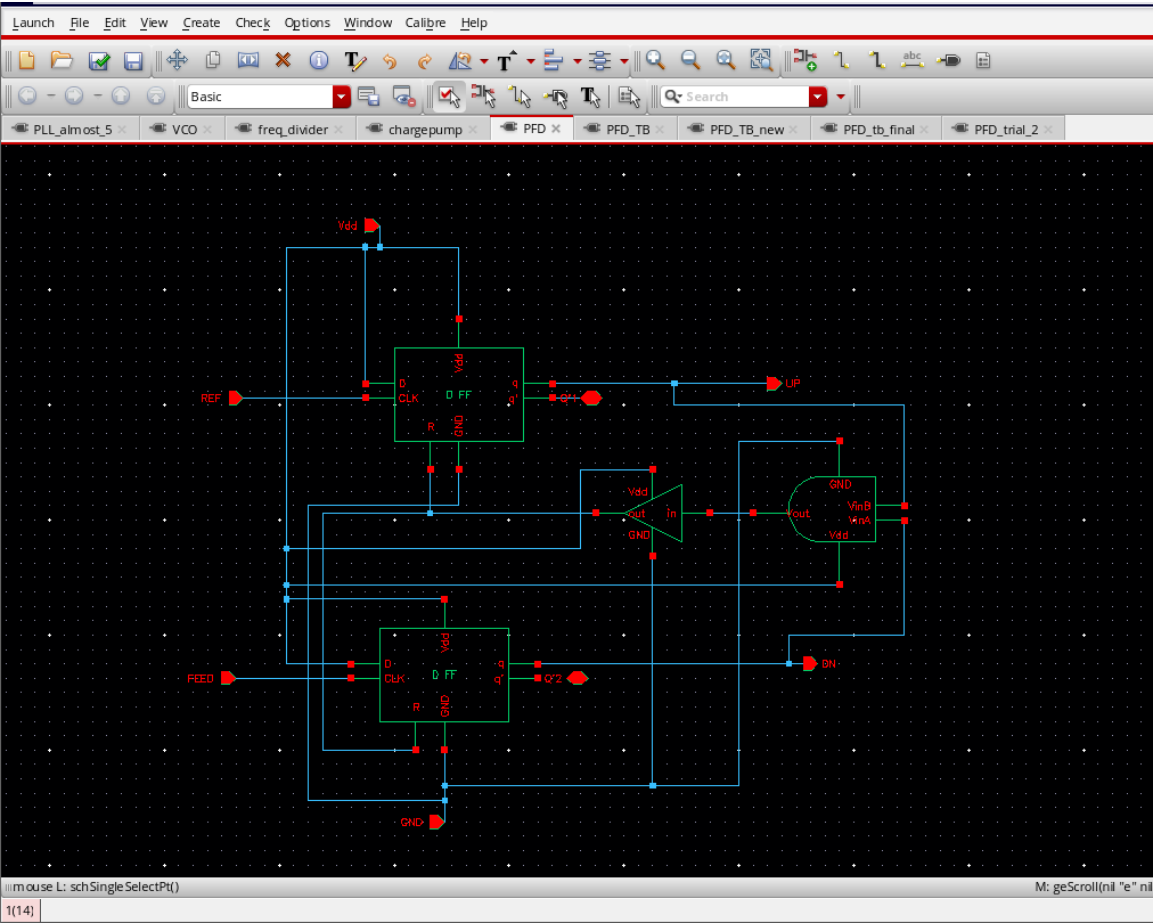
Design Blocks

- PFD
- Charge Pump
- Loop Filter
- 5-Stage Current-Starved Ring VCO
- Divide-by-8 Frequency Divider
- Complete PLL

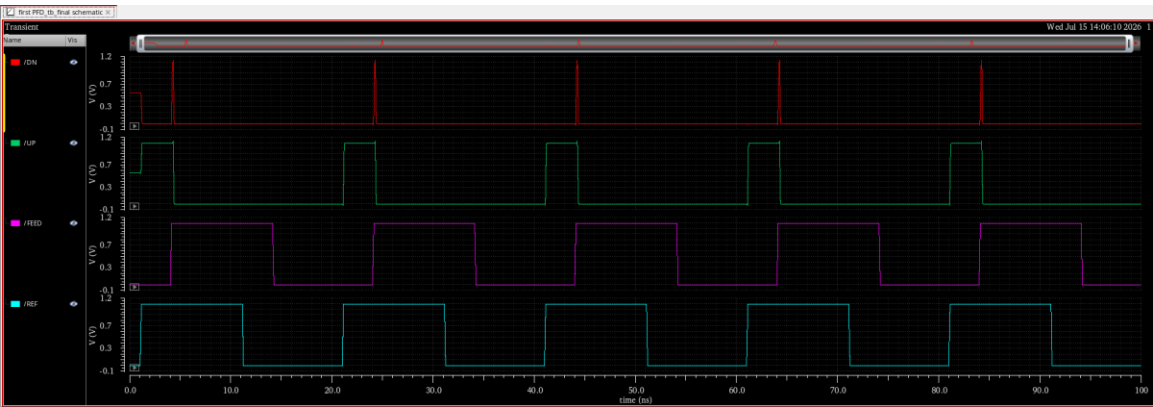
Phase Frequency Detector (PFD)

The Phase Frequency Detector is the brain of the PLL. It compares the phase and frequency of the reference signal with the feedback signal and generates UP and DOWN control pulses. These pulses indicate whether the VCO frequency needs to increase or decrease.

PFD Schematic



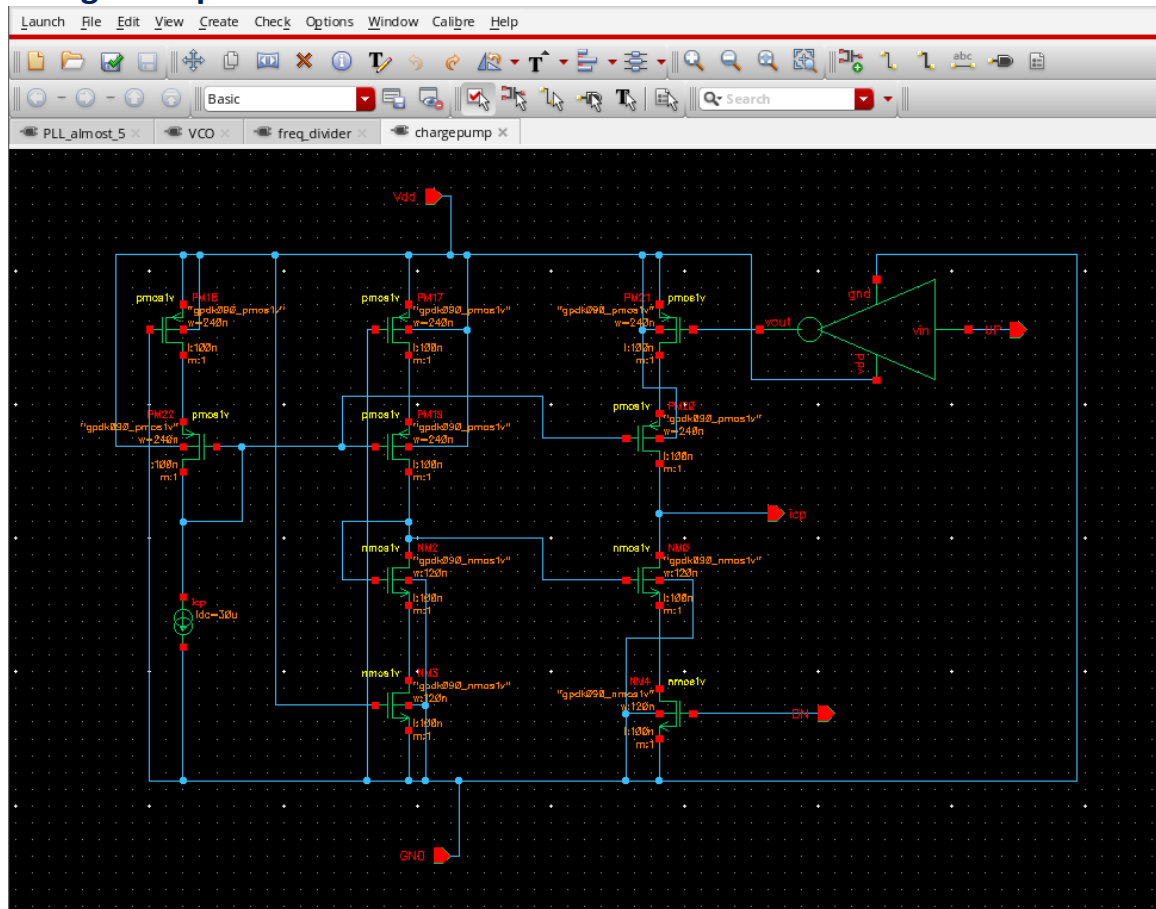
PFD Output



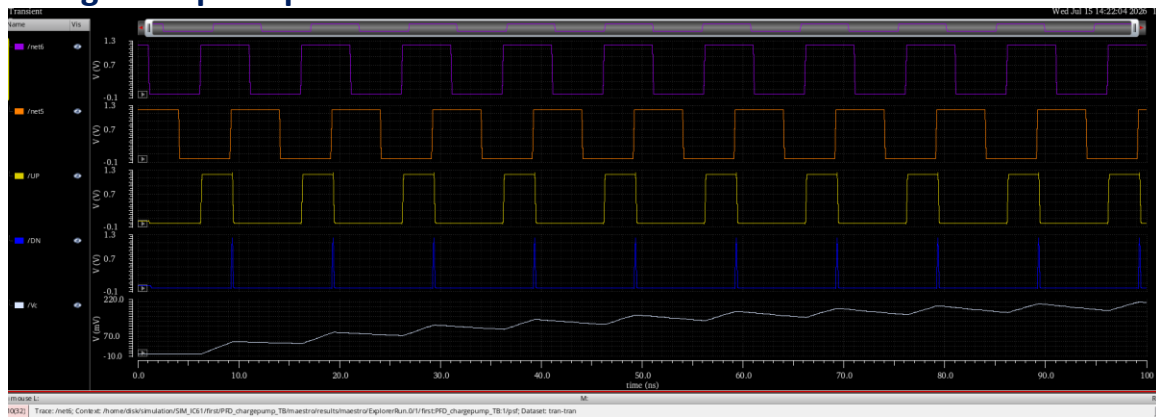
Charge Pump (CP)

The Charge Pump acts as a current steering circuit. Based on the UP and DOWN pulses from the PFD, it injects or removes current from the loop filter, converting digital timing information into an analog control signal.

Charge Pump Schematic



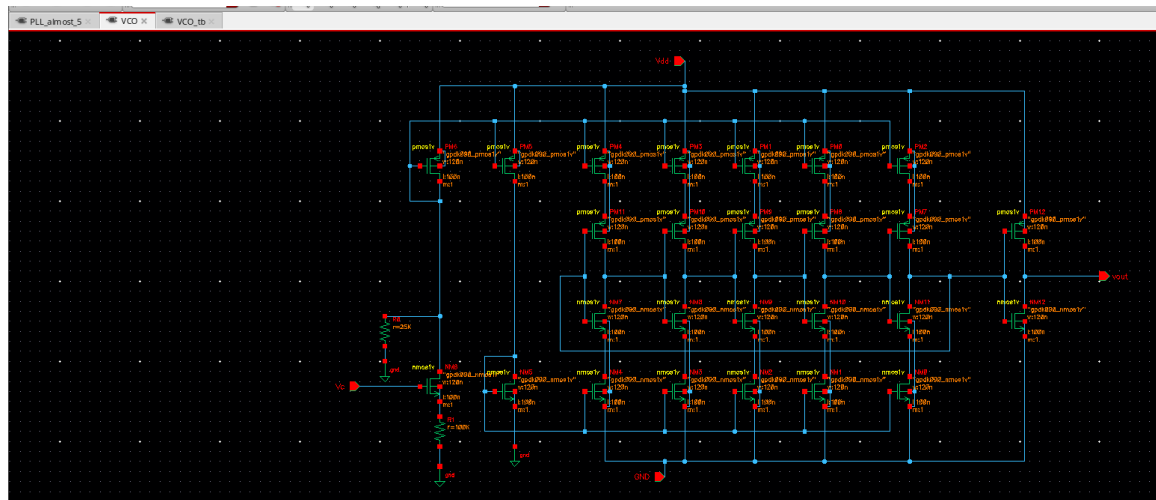
Charge Pump Output



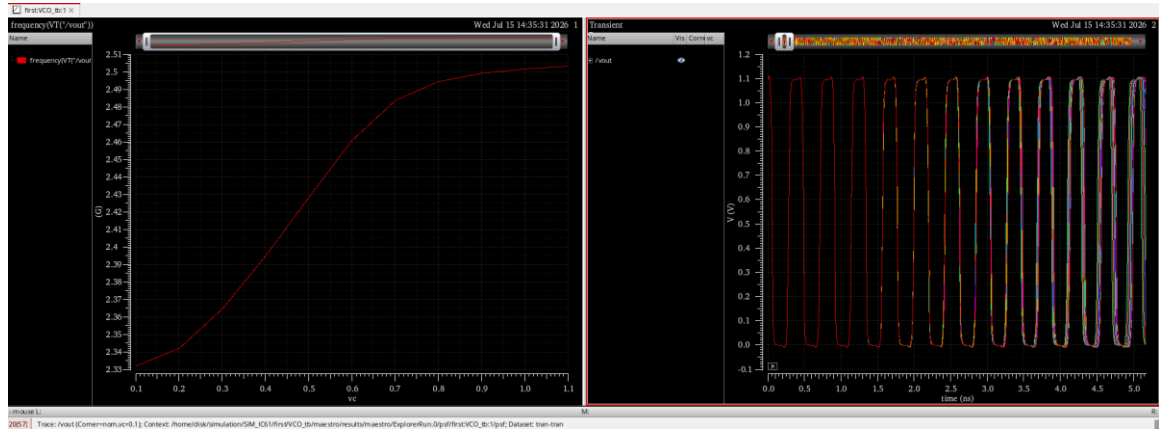
Voltage Controlled Oscillator (VCO)

The Voltage Controlled Oscillator converts the control voltage into a high-frequency oscillating signal. In this project, a 5-stage Current-Starved Ring Oscillator is used, where the oscillation frequency varies according to the applied control voltage.

VCO Schematic



VCO Oscillation



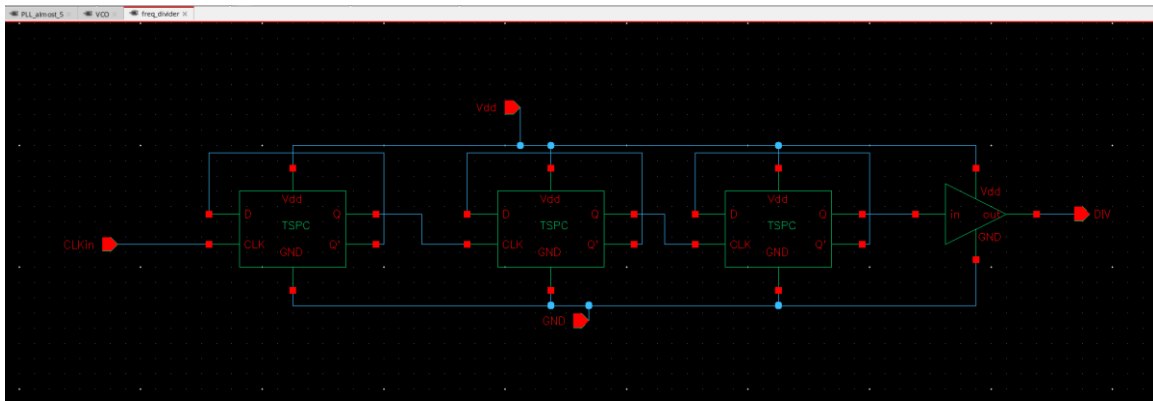
Frequency vs Vc and Kvco Plot

frequency(VT("/vout")) deriv(frequenc... ✕				
	vc	frequen...er nom	vc	deriv(fr...ner nom
1	100.0E-3	2.332E9	200.0E-3	99.49E6
2	200.0E-3	2.342E9	300.0E-3	227.7E6
3	300.0E-3	2.365E9	400.0E-3	302.4E6
4	400.0E-3	2.395E9	500.0E-3	330.3E6
5	500.0E-3	2.428E9	600.0E-3	327.6E6
6	600.0E-3	2.461E9	700.0E-3	230.2E6
7	700.0E-3	2.484E9	800.0E-3	106.3E6
8	800.0E-3	2.495E9	900.0E-3	47.81E6
9	900.0E-3	2.500E9	1.000	25.33E6
10	1.000	2.502E9	1.100	14.63E6
11	1.100	2.504E9		

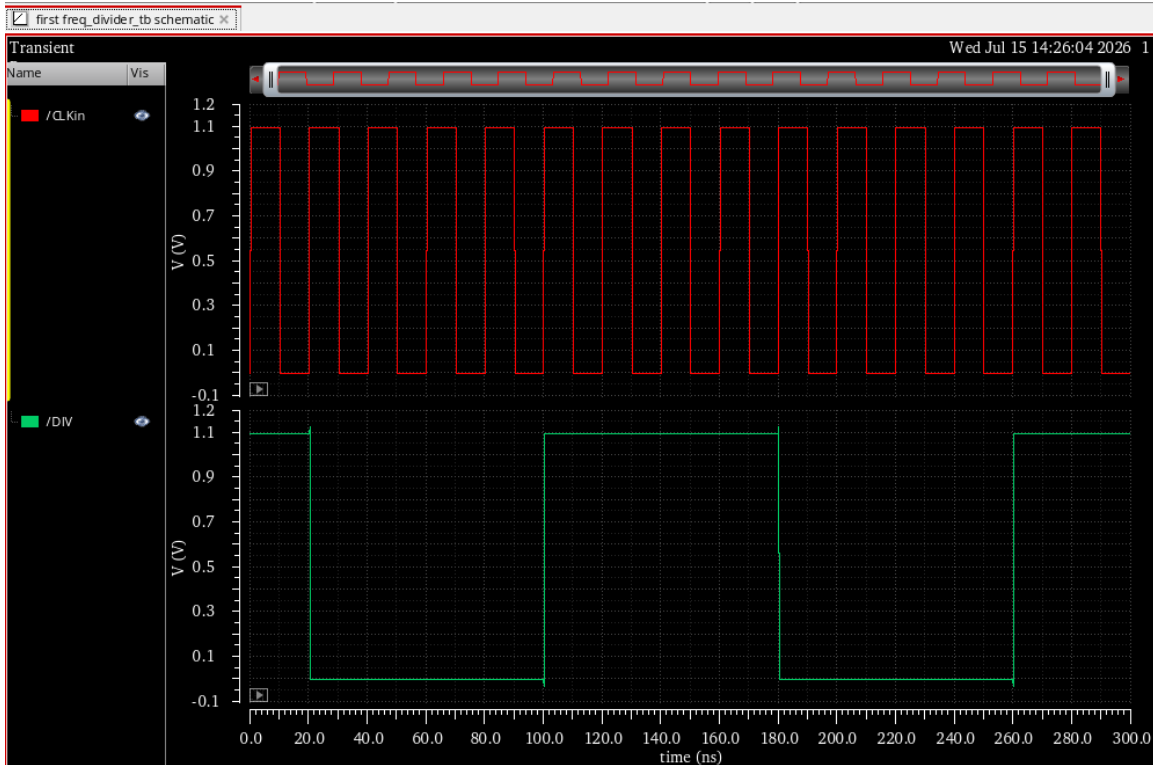
Frequency Divider

The Frequency Divider scales down the VCO output frequency before feeding it back to the PFD. It enables the PLL to generate output frequencies that are integer multiples of the reference frequency while maintaining phase synchronization.

Divider Schematic



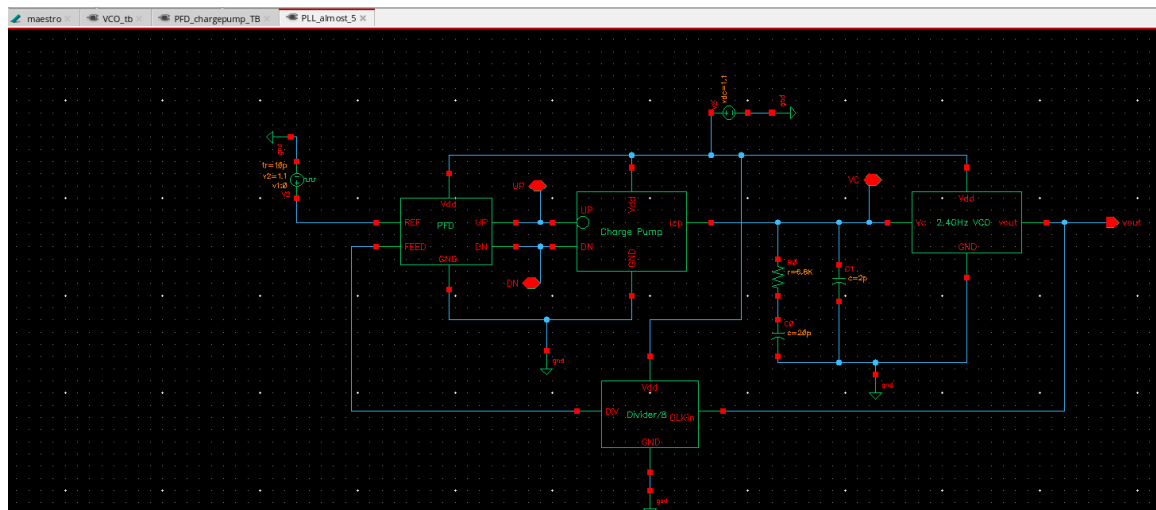
Divider Output



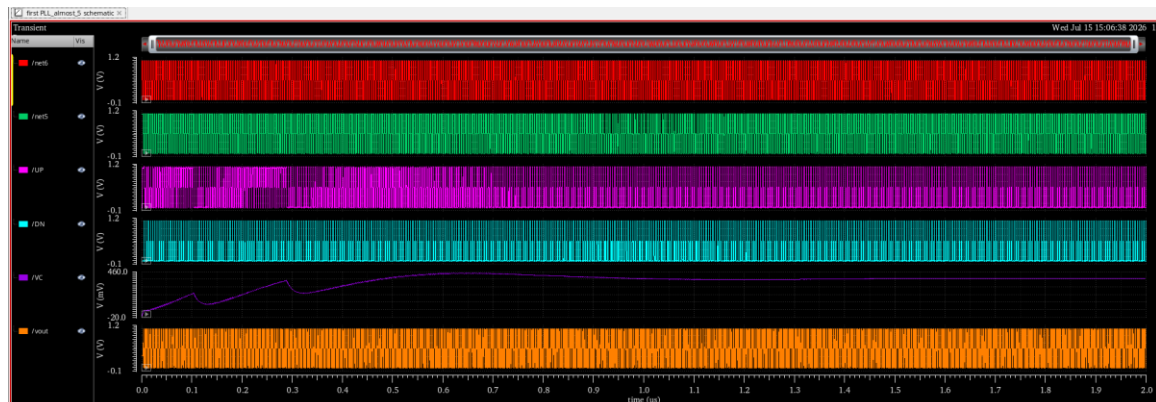
Complete PLL

The complete Charge Pump PLL integrates the PFD, Charge Pump, Loop Filter, VCO, and Frequency Divider into a closed feedback system. The loop continuously adjusts the VCO until the feedback signal matches the reference signal in both phase and frequency, achieving a stable locked condition.

PLL Schematic



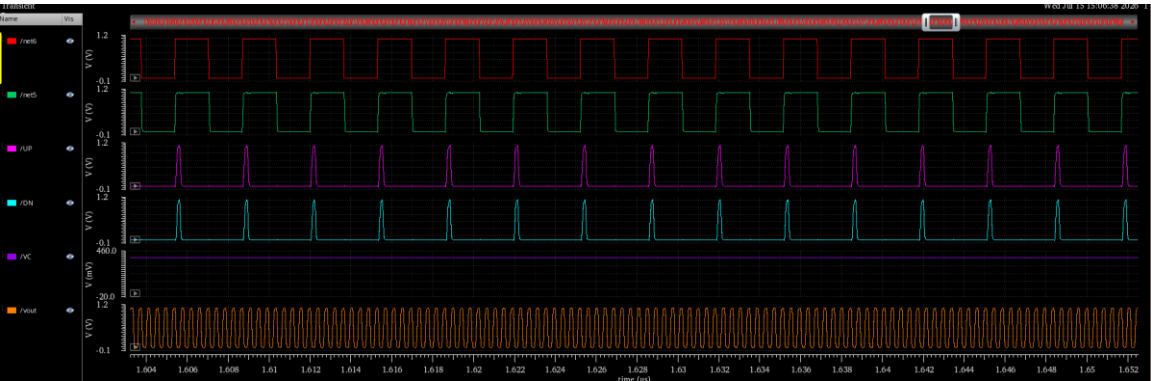
PLL Lock Waveform



Frequency of PLL

frequency(VT("/vout")) ×	
Expression	Value
frequency(VT("/vout"))	2.410E9

Reference vs Feedback



Results

Performance Parameter	Result
Technology	GPDK090
Supply Voltage (VDD)	1.2V
PLL Architecture	Charge Pump PLL
VCO Type	5-Stage Current-Starved Ring Oscillator
Charge Pump Current (Icp)	30uA
Loop Filter	Passive second order

R	8.8 K Ω
C1	20pF
C2	2pF
Divider ratio	Divide by 8
Reference Frequency	301.25MHz
Nominal PLL Output Frequency	2.41GHz
VCO Frequency Range	2.332-2.504 GHz
Average K_{VCO}	172 MHz/V
Peak K_{VCO}	$\approx$ 330 MHz/V
Lock Status	Successfully Achieved

Conclusion

A complete transistor-level Charge Pump PLL was successfully designed and simulated using Cadence Virtuoso in GPDK090 technology. Individual blocks including the PFD, Charge Pump, Loop Filter, Current-Starved Ring VCO, and Divide-by-8 Frequency Divider were verified independently before integration into the complete PLL. The implemented design achieved a nominal locked output frequency of approximately **2.41 GHz** with an average K_{VCO} of **172 MHz/V**, demonstrating stable frequency synthesis and successful phase locking.